

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –**
SIH26034
- **Problem Statement Title –**
AI-Powered Automated Inspection & Enforcement System for
Legal Metrology (Packaged Commodities) Rules
- **Theme –**
Smart Automation / Regulatory Technology & Governance
- **PS Category –**
Software
- **Team ID –**
[Registered Portal Team ID]
- **Team Name (Registered on portal) –**
The Null Pointers

PRAMAAN



*National Legal Metrology Compliance &
Regulatory Surveillance Operating System*

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- ✓ EAN-13 Optical Metrology ($\pm 0.1\text{mm}$)
 - ✓ Rule 7(2) & Rule 6 Deterministic Logic
 - ✓ Section 65B Admissible PDF Notices
 - ✓ Hardware Motion Guard ($< 0.18\text{g}$)
 - ✓ 24/7 E-Commerce Surveillance

PRAMAAN (■■■■■■■■)

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed Explanation

- **Unified Regulatory OS:** Automates end-to-end statutory enforcement of Legal Metrology (Packaged Commodities) Rules, 2011 across retail and e-commerce.
- **3-Tier Ecosystem:** Mobile Field Scanner (AR HUD), Web Enforcement Portal (Supervisory queue & notices), and Autonomous Market Surveillance Scraper.
- **Deterministic Logic:** Replaces contestable black-box AI with auditable, mathematical rules codification of statutory provisions.
- **Evidence-Grade Notice:** Automatically synthesizes Section 65B Indian Evidence Act compliant PDF notices with cryptographic SHA-256 hashes.

• How It Addresses the Problem

- **Eliminates Physical Calipers:** Solves the slow, manual raid measurement bottleneck by measuring font heights to $\pm 0.1\text{mm}$ in < 4 seconds.
- **Defeats Deceptive Packaging:** Checks pack weights against Second Schedule permissible sizes, catching subtle shrinkflation instantly.
- **Replaces Subjective Disputes:** Generates tamper-evident photographic bounding-box exhibits that withstand judicial scrutiny in consumer courts.
- **Monitors E-Commerce at Scale:** Autonomous scraper audits thousands of listings on Blinkit, Zepto, Amazon, and Flipkart 24/7.
- **Eliminates False Accusations:** Dual-tier confidence gate ($< 85\%$) diverts edge cases to human officer review.

• Innovation and Uniqueness

- **Universal Optical Ruler:** World-first metrology utilizing standard GS1 EAN-13 barcodes (nominal 37.29mm) as physical calibration markers.
- **Hardware Motion Guard:** Integrates device accelerometer sensors (jitter $< 0.18\text{g}$) to mathematically block motion-blurred captures.
- **Human-in-the-Loop Triage:** Automated enforcement for high confidence ($> 85\%$); supervisory officer queue for complex edge cases.
- **True Offline Resilience:** Encrypted local SQLite queue enables complete field raid inspections in remote markets with zero cellular network.

TECHNICAL APPROACH

• Technologies to be Used

- **Mobile Scanner:** React Native (Expo 50), TypeScript, CameraX HUD, Accelerometer Sensor API, Encrypted SQLite Queue.
- **Web Enforcement Portal:** React 18, Vite, TailwindCSS, TypeScript, Lucide Icons, Axios Client (JWT Bearer auth).
- **Backend Gateway:** Python 3.11, FastAPI (ASGI async), Pydantic v2, SlowAPI rate-limiting, Structlog JSON logging.
- **Computer Vision:** OpenCV (Laplacian blur Var < 100, Specular Glare > 5%, WCAG Luminance Contrast Ratio > 3.0).
- **Optical Metrology:** PyZbar (EAN-13 37.29mm calibration standard, Scale factor $S = 37.29 / W_{px}$).
- **OCR & VLM Engine:** PaddleOCR Bilingual (English + Hindi) + Google Gemini 1.5/2.5 Flash VLM fallback + Regex Normalizer.
- **Database & Persistence:** PostgreSQL 16 (Scans & Audit logs), MinIO / S3 content-addressed storage.
- **Court Admissibility:** Jinja2 + WeasyPrint PDF/A with SHA-256 tamper-evident digital certificate chains.

• Methodology & Implementation Pipeline

- **1. Ingestion:** Captures label via Mobile AR HUD, Bulk Web Upload, or E-Commerce Scraper; generates instant SHA-256 hash.
- **2. Preprocessing & Quality Check:** Evaluates blur and glare; PyZbar detects EAN-13 barcode to compute scale factor: $S = 37.29\text{mm} / W_{px}$. Measures numeral height: $H_{mm} = H_{px} * S$.
- **3. Bilingual Extraction:** PaddleOCR extracts structured text; Gemini VLM handles curved/degraded packaging; regex parses Net Qty, MRP, Mfg Date, Origin.
- **4. Rules Evaluation:** Codified engine verifies Rule 7(2) Table I (font height vs PDP), Rule 6 (mandatory declarations), and Second Schedule (standard sizes).
- **5. Confidence Triage:** Score $\geq 85\%$ authorizes automatic violation notice; Score < 85% routes scan to Officer Review Queue for judicial safety.
- **6. Enforcement Notice PDF:** Renders Section 65B compliant legal notice with visual bounding boxes, officer badge, and immutable SHA-256 audit trail.

WORKING PROTOTYPE: Fully functional & cloud-deployed; 34/34 automated pytest tests passing; sub-4s later

FEASIBILITY AND VIABILITY

• Analysis of Feasibility

- **Zero Hardware Capex:** Runs on any standard Android/iOS smartphone already used by field inspectors; zero specialized calipers needed.
- **Economic Viability:** Built on 100% open-source software (Python, PostgreSQL, React). Cloud compute cost is < ₹0.20 per raid inspection.
- **Legal Admissibility:** Designed under Section 65B of Indian Evidence Act / BSA 2023 with cryptographic SHA-256 custody.
- **High Scalability:** Stateless FastAPI microservices horizontally scale across 28 states & 8 UTs on cloud container clusters.
- **Immediate Utility:** Direct operational utility for retail raids, border customs import checks, and e-commerce compliance.

• Potential Challenges & Risks

- **Cylindrical Packaging Curvature:** Bottles, cans, and deformable pouches cause perspective distortion of text and barcode widths.
- **Specular Glare & Reflections:** Shiny foil laminates in wholesale markets cause reflections that obscure numeral contours.
- **Rural Network Blackouts:** Field raids in basement godowns or rural mandis lack active 4G/5G cellular connectivity.
- **Adversarial Formatting:** Manufacturers deliberately using stylized typography, poor contrast, or buried declarations.
- **Judicial Scrutiny:** Consumer courts demanding absolute proof of measurement accuracy.

• Strategies for Overcoming

- **Cylindrical Unwrapping:** Perspective transformation and cylindrical surface unrolling algorithms normalize curved coordinates.
- **CLAHE & Glare Rejection:** OpenCV specular glare mask rejects highlights >5% and applies adaptive histogram equalization.
- **Encrypted Offline Queue:** Local SQLite/AsyncStorage caches scans offline and automatically syncs when network returns.
- **Dual OCR + VLM Pipeline:** PaddleOCR handles clean text; Gemini VLM acts as fallback for stylized, folded, or multilingual labels.
- **Human-in-the-Loop Triage:** Mandatory supervisory officer review for any scan with extraction confidence < 85%.

IMPACT AND BENEFITS

• Potential Impact on Target Audience

- **Legal Metrology Field Inspectors:** Reduces physical inspection time from 15-20 minutes to under 4 seconds per commodity, multiplying daily raid throughput by 10x-15x.
- **Department of Consumer Affairs:** Provides centralized national surveillance dashboard across 28 states with real-time heatmaps of habitual non-compliant brands.
- **1.4 Billion Indian Consumers:** Protects citizens against deceptive undersizing ('shrinkflation'), missing manufacturer origins, expired batches, and hidden tax surcharges.
- **E-Commerce Marketplaces:** Enables autonomous regulatory pre-screening of 3rd-party sellers before product listings go live on consumer apps.
- **Compliant FMCG Manufacturers:** Fosters a fair and standardized compliance ecosystem free from arbitrary officer subjectivity and inspector bias.

• Benefits of the Solution (Social, Economic, Environmental)

- ❖ **Economic Benefits:**
 - **Revenue Recovery:** Recovers crores of rupees in compounding fees and penalties from chronic violators.
 - **Anti-Shrinkflation:** Prevents deceptive 10-15% net quantity reductions by enforcing Second Schedule sizes.
- ❖ **Social & Public Health Benefits:**
 - **Consumer Safety:** Guarantees mandatory legibility of expiry dates, allergen warnings, and consumer care on baby foods and pharmaceuticals.
 - **Swift Justice:** Court-admissible notices resolve consumer court proceedings in days rather than years.
- ❖ **Environmental & Governance Benefits:**
 - **100% Paperless Raids:** Eliminates manual paper case diaries and carbon-copy seizure memos with encrypted digital records.
 - **Corruption-Proof:** Immutable cryptographic audit trails prevent record tampering or unauthorized suppression.

RESEARCH AND REFERENCES

• Statutory & Legal References

- **1. Legal Metrology Act, 2009 (Act No. 1 of 2010):** Ministry of Consumer Affairs, Food & Public Distribution, Government of India. Mandates standard weights, measures, and retail declarations.
- **2. Legal Metrology (Packaged Commodities) Rules, 2011 (amended up to 2022):** Specific codification of Rule 6 (Mandatory Declarations), Rule 7(2) Table I & II (Numeral Heights), Rule 26 (Small Package Exemptions).
- **3. Second Schedule of LMPC Rules, 2011:** Permissible standardized packaging quantities for foodstuff, personal care, and lubricants.
- **4. Section 65B of Indian Evidence Act, 1872 / Bharatiya Sakshya Adhiniyam, 2023:** Special provisions regarding admissibility of electronic records, SHA-256 digital hashes, and officer certification.
- **5. Consumer Protection (E-Commerce) Rules, 2020:** Mandatory display of country of origin, MRP, and net quantity on e-commerce platforms.

• Technical Standards & Project Links

- **1. GS1 General Specifications (Release 23.0):** Section 5.2.1: EAN-13 Barcode Symbol Specifications. Establishes the universal nominal barcode width ($W = 37.29$ mm) for physical scale calibration.
- **2. ISO/IEC 15420:2009 Standard:** Information technology — Automatic identification and data capture techniques — EAN/UPC bar code symbology specification.
- **3. Du, Y., et al. (2020) 'PP-OCR: A Practical Ultra Lightweight OCR System':** arXiv:2009.09941. Bilingual text detection and recognition methodology.
- **4. W3C WCAG 2.1 Contrast Guidelines:** Relative luminance contrast ratio standard (3:1 minimum compliance threshold for package legibility).
- **5. Live Working Codebase & Deployment:**
 - GitHub Repository: <https://github.com/Debddi/Pramaan>
 - Includes tested FastAPI backend, React Native mobile app, web dashboard, 34 automated unit tests, and 1-click cloud deployment specs.